Cover letter

27 July 2026

Re: Submission of "Calibration-derived decoder discriminability is associated with online P300-speller accuracy, but the fitted mapping does not transport across cohorts" as a Paper

Dear Editors of *Journal of Neural Engineering*,

We submit "Calibration-derived decoder discriminability is associated with online P300-speller accuracy, but the fitted mapping does not transport across cohorts" for consideration as a Paper in *Journal of Neural Engineering*.

Submission and journal fit. A calibration-derived score has repeatedly been related to online P300-speller accuracy within the cohort where it was measured, including by Mainsah and colleagues in this journal (*J Neural Eng* 2016;13:066007), who derived speller accuracy analytically from a calibration-derived detectability index. Whether a mapping fitted in one set of cohorts transports to a cohort it has never seen is the question that decides whether such a score can be reported anywhere other than where it was developed, and it is the question this manuscript answers.

What is new. Using the public BigP3BCI archive, we evaluated a calibration-derived score across 18 source-study cohorts with a leave-one-study-out design (271 participants, 739 session-condition records, 19,611 selections), rather than the single-cohort evaluations the literature has reported to date. Each withheld cohort's expected accuracy was estimated purely from the other 17, with no information from that cohort used in fitting the mapping evaluated against it.

Principal results. The association held in every withheld cohort (participant-level r = 0.714) but the fitted mapping from score to expected accuracy did not transport: the calibration intercept had a between-cohort standard deviation of tau = 0.87, spanning a 95% interval for an unrepresented cohort of -1.97 to 1.85 on the log-odds scale, and the slope agreed at tau = 0.43. Both conclusions hold at the lower confidence limit of tau, so the finding does not depend on the number of cohorts being read favourably. A four-cohort subgroup limited to the documented ALS cohorts alone, evaluated on its own, gave a materially tighter and more favourable picture (uncorrected between-cohort slope spread 0.223, against 0.560 across all 18). This is a concrete demonstration of how few-cohort evaluation understates real-world variability.

Why this matters to your readership. The manuscript separates two claims the field has tended to report together: an association that appears in every cohort examined, and a calibrated mapping that transports between them. A calibration score can support ranking sessions within a setting and can support a data-quality screen; it should not be used to report an expected accuracy in a cohort where the mapping was not developed, without local recalibration. This is a transportability evaluation of a widely proposed relationship, not a classifier-improvement study.

Declarations. The authors declare no competing interests. This work has not been submitted elsewhere and is not under consideration by any other journal.

Sincerely,

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